

COURSE SYLLABUS

MLOps: Machine Learning Operations

Language of Instruction: English

5 Weeks (20 Hours)

Course instructor: TRAN Cong Minh

Course Description

This course develops the engineering practices for turning a machine learning experiment into a reproducible, testable and maintainable service. It covers data preparation, tracked baseline training, containerised prediction services, automated checks, monitoring and failure recovery. The emphasis is reliable systems and justified engineering decisions, rather than model complexity or predictive performance alone.

Lectures support a cumulative team project. In guided labs, students implement each stage, inspect actual execution results and explain their decisions.

Course Objectives

Upon successful completion of this course, students will be able to:

- Manage source code and data under version control and reproduce a tracked baseline model.
- Develop a tested prediction API and deploy it as a containerised application.
- Design automated training pipelines that incorporate evaluation gates and continuous integration.
- Monitor a machine learning service in operation, diagnose incidents and verify recovery using logs, tests and model-version checks.
- Release versioned models, perform controlled rollbacks and produce documentation suitable for engineering handover.

Course Project

Students work in teams of three or four to develop a service that estimates hourly bicycle rental demand from calendar and weather variables, using the [UCI Bike Sharing dataset](#). The project is developed incrementally through the weekly laboratory sessions, progressing from a tracked baseline model to a tested API, automated quality checks, operational monitoring and documented recovery procedures.

Teams submit weekly checkpoint tag links `m1–m4`, with brief check results and blockers. The final submission is a GitHub Release `v1.0.0` linking the code revision, configuration, trained model and verification evidence, with a concise README and recovery runbook.

Technical stack: Git, Python with uv, scikit-learn, MLflow, FastAPI, pytest, Docker and GitHub Actions.

Assessment

The final grade is determined by two components:

- **Team project and presentation (60%).** The final system is evaluated on its reproducibility, the testing of the API and container, the pipeline gates and continuous integration configuration, structured logging, incident diagnosis and verified recovery.
- **Individual laboratory reports (40%).** Four reports in Weeks 1–4 each contribute 10% of the course grade, due the following Sunday at 23:59 in the published time zone. Each records contributions, code/PR and execution evidence, problems/resolutions and learning. Receipt is checked weekly; all four snapshots are graded at course end for correctness, evidence and understanding, clarified by individual presentation answers. No Week 5 report is required.

Course Outline

Week	Lectures	Labs	Required submissions
Week 1	Lecture 1 Foundations of Machine Learning Operations Lecture 2 Data Quality for Reproducible Baselines	Lab 1 Shared Repository, Git Workflow & Code Quality Checks Lab 2 Data Validation, Chronological Split & Baseline Tracking	Individual (10%) reports/week-1.md Team checkpoint m1 Tag link and short note: PR/check results; validated split, tracked baseline, checksum, configuration and metrics.
Week 2	Lecture 3 Reproducible Environments with Containers Lecture 4 Prediction Services with Automated Testing	Lab 3 Dependency Locking, Docker Build & Clean-Environment Rerun Lab 4 Prediction API, Input Validation & Containerised Tests	Individual (10%) reports/week-2.md Team checkpoint m2 Tag link and short note: dependency lock, clean rerun, working API/container and test results.
Week 3	Lecture 5 Deploying Models as Observable Services Lecture 6 Continuous Integration for Machine Learning Projects	Lab 5 Service Deployment, Structured Logging & Model Version Checks Lab 6 GitHub Actions, CI Checks & Regression Tests	Individual (10%) reports/week-3.md Team checkpoint m3 Tag link and short note: deployed service, structured logs, model-version checks and failed/repaired CI evidence.
Week 4	Lecture 7 Automated Training Pipelines with Evaluation Gates Lecture 8 Incident Response for Deployed Models	Lab 7 Training Pipeline, Model Evaluation & Quality Gate Lab 8 Service Monitoring, Incident Diagnosis & Rollback Verification	Individual (10%) reports/week-4.md Team checkpoint m4 Tag link and short note: pipeline/gate decisions, incident diagnosis and verified rollback.
Week 5	Project clinic, final verification & team project presentations		Individual No weekly report. Final team submission (60%) GitHub Release v1.0.0: code/tests, model, verification evidence, README/runbook and release notes; group presentation.

Submission notes

Weeks 1–4 team tags are ungraded checkpoints, not weekly GitHub Releases. Submission snapshots are retained; dates and collection details are published separately.